

LANE DEPARTURE WARNING SYSTEM

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ABSTRACT

KEYWORDS: Traffic Safety, LDWS, Image Processing, Hough Transform, Canny Edge Detector, Euclidean Distance

Today, one of the largest areas of research and development in the automobile industry is road safety. Many deaths and injuries occur every year on public roads from accidents caused by sleepy drivers, that technology could have been used to prevent. In Advance Driver Assistance system (ADAS), lane detection is an essential component. The Lane Departure Warning System (LDWS) is a safety system that helps to prevent accidents. Drivers can easily become distracted and lose full focus on the road, especially at the end of a long and tiring working day, therefore LDWS helps drivers to remain alert. LDWS assesses where the vehicle is driving and detects the lines on the road. If the vehicle is in danger of leaving the lane, this triggers a warning, helping the drivers to remain alert. The system takes into account whether the direction indicator is being used, thereby preventing false alarms when a driver is intentionally changing lane. It uses the principle of Hough transform and Canny edge detector to detect lane lines from real-time camera images fed from the front-end camera of the vehicle. The main purpose of the system is to detect the lane departure to avoid road accidents and to provide safety for pedestrians. The proposed method uses the piecewise stretching function to set the contrast level of the road images. The region of interest is divided into two segment namely left and right sub-region using segmentation. After pre-processing the ROI, lane detection takes place individually in each sub-region using the Hough transform algorithm. The lane departure identification is based on the Euclidean distance measure. If the driver tends to change the lane then the warning is issued to the driver. One of the challenging task of the project is to correctly detect the road lanes in various environmental conditions. This feature relies on painted lane markings to operate and it is not designed to work with markers that are faded, covered, in disrepair, or are overly complicated.

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ABBREVIATIONS

ADAS	Advance Driver Assistance System
LDWS	Lane Departure Warning System
ROI	Region of Interest
IDLE	Integrated Development and Learning Environment
RGB	Red Green Blue
HT	Hough transform
RANSAC	Random Sample Consensus
CNN	Convolutional Neural Network

CHAPTER 1

INTRODUCTION

In today's world, there are numerous issues concerning the common motorist and the safety of roadside construction workers who have to operate in close proximity to them. Things like negligent driving and speeding pose some high concerns and threats to us humans. Recommendations were evaluated and examined to make in terms of having a device with the ability to monitor roadside construction workers as well as ongoing traffic.



Figure 1.1: Traffic congestion on Kerala roads

According to the latest data available from the Ministry of Road Transport and Highways in India, there were a total of 4,49,002 road accidents in the country in 2019, which translates to an average of 1,229 accidents per day. These accidents resulted in 1,54,732 deaths and 4,51,361 injuries. Also our Kerala had a total of 39,986 road accidents in 2019 which means an average of around 109 accidents per day in the state. These accidents resulted in 4,267 deaths and 40,743 injuries. It's important to note that this data is from 2019 and the number of accidents may have fluctuated in subsequent years.

The present invention relates to an Automatic Driver Assistance Systems (ADAS) that can provide drivers an assistance during their car drives. It refers to a collection of advanced technologies and features designed to assist drivers in operating vehicles

more safely and efficiently. ADAS technologies use sensors, cameras, radar, and other devices to gather information about the vehicle's surroundings and provide feedback or automated interventions to enhance the driving experience.

In crowded road environments such as commute and commute times, vehicles that try to enter the intersection first often interfere with other lanes.

1.1 Types of ADAS

ADAS technologies are continually advancing, and newer vehicles may include additional features like pedestrian detection, lane-keeping assist, drowsiness monitoring, and more. However, it's important to note that ADAS systems are designed to assist drivers and should not be relied upon as a substitute for attentive driving. Drivers should always remain focused and engaged while operating a vehicle, even with these advanced features. Here are some common examples of ADAS features:

- Adaptive Cruise Control (ACC): ACC automatically adjusts the vehicle's speed to maintain a safe distance from the vehicle ahead. It uses radar or laser sensors to detect the distance and relative speed of other vehicles.
- Lane Departure Warning (LDW): LDW alerts the driver if the vehicle unintentionally drifts out of its lane without using the turn signal. It uses cameras or sensors to monitor lane markings on the road.
- Forward Collision Warning (FCW): FCW warns the driver if it detects an imminent collision with the vehicle ahead. It uses sensors or cameras to monitor the distance and speed of the vehicle in front.
- Automatic Emergency Braking (AEB): AEB automatically applies the brakes if a potential collision is detected and the driver fails to respond in time. It helps reduce the severity of accidents or avoid them altogether.
- Blind Spot Detection (BSD): BSD uses sensors to monitor the vehicle's blind spots and alerts the driver if another vehicle is detected in those areas. It helps prevent collisions while changing lanes.
- Parking Assistance Systems (PAS): These systems help drivers maneuver the vehicle into parking spaces. They can include features like parking sensors, rearview cameras, and automated steering.
- Traffic Sign Recognition (TSR): TSR uses cameras or sensors to detect and read traffic signs such as speed limits, stop signs, or no-entry signs. It provides visual or audible alerts to help the driver stay informed about the current road conditions.

Before the advent of ADAS technologies, drivers relied on traditional methods and their own skills to assist them while driving. Here are some traditional methods commonly used:

- **Mirrors and Blind Spot Checks:** Drivers would use their rearview and side mirrors to monitor the vehicles behind and beside them. They would also perform shoulder checks or glance over their shoulders to check blind spots before changing lanes or making turns.
- **Turn Signals:** Drivers would manually activate their turn signals to indicate their intention to change lanes or make turns. This would alert other drivers of their intended actions.
- **Speed Control:** Drivers would manually control their vehicle's speed by adjusting the accelerator and brake pedals based on the road conditions, traffic, and speed limits.
- **Manual Braking:** In the event of a potential collision, drivers would rely on their reflexes and judgment to apply the brakes manually to avoid or minimize the impact.
- **Manual Steering:** Drivers would use the steering wheel to control the direction of their vehicle, ensuring it stays within the lane and follows the road's curvature.

These traditional methods require drivers to be attentive, skilled, and make split-second decisions based on their observations and experience. While ADAS technologies have added an extra layer of assistance, it is still important for drivers to be aware of these traditional methods and practice safe driving habits.

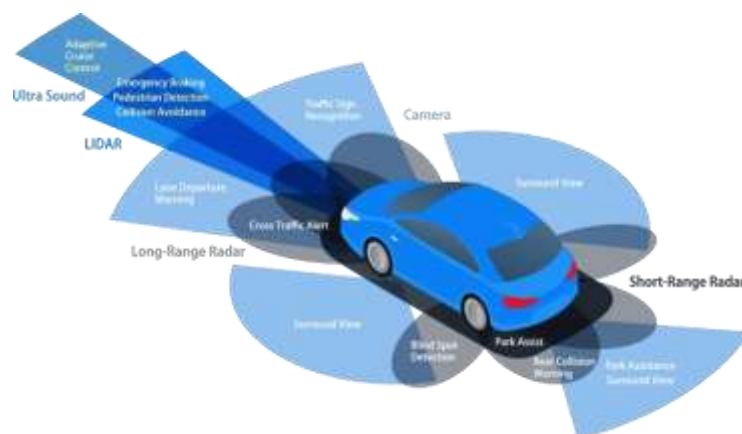


Figure 1.2: Different types of ADAS

1.1.1 Advantages and Disadvantages of ADAS

Automatic Driver Assistance Systems offer several advantages and disadvantages. Let's explore them:

Advantages of ADAS

1. **Enhanced Safety:** ADAS technologies help improve overall road safety by providing assistance and alerts to drivers. Features like automatic emergency braking, lane departure warning, and blind spot detection can help prevent accidents and reduce the severity of collisions.
2. **Collision Mitigation:** ADAS systems can detect potential collisions or hazards and take proactive measures to mitigate them. They can apply the brakes, adjust vehicle speed, or provide warnings to the driver, reducing the risk of accidents.
3. **Improved Driver Assistance:** ADAS technologies can assist drivers by reducing driver fatigue and workload. Features like adaptive cruise control and lane-keeping assist can help maintain a safe distance from other vehicles and keep the vehicle centered within the lane, respectively.
4. **Increased Convenience:** ADAS features can provide convenience to drivers. Parking assistance systems, for example, make it easier to park in tight spaces. Traffic sign recognition can help drivers stay informed about speed limits and other important traffic regulations.
5. **Potential for Fuel Efficiency:** Some ADAS technologies, such as adaptive cruise control, can optimize fuel consumption by maintaining a consistent speed and adjusting it based on traffic conditions. This can lead to improved fuel efficiency.

Disadvantages of ADAS

1. **Overreliance and Complacency:** One of the major concerns with ADAS technologies is the potential for drivers to become overly reliant on them. This may lead to complacency and reduced attentiveness, as drivers may assume the systems will handle all situations perfectly. However, ADAS systems have limitations, and drivers should always remain actively engaged and ready to take control.
2. **False Alerts and Malfunctions:** ADAS systems can sometimes generate false alerts or experience malfunctions. For example, lane departure warning systems may mistake faded lane markings for valid ones, resulting in unnecessary warnings. Technical malfunctions can also occur, impacting the accuracy and reliability of the systems.
3. **High Cost:** ADAS technologies typically add to the cost of vehicles. The advanced sensors, cameras, and processing systems required for these features can increase the purchase price, making them more expensive compared to vehicles without ADAS.
4. **Potential Learning Curve:** Some ADAS features may require drivers to familiarize themselves with new technologies and adapt to their functionality. This learning curve may take some time, and drivers may need to adjust their driving habits accordingly.
5. **Limited Effectiveness in Certain Conditions:** ADAS systems may have limitations in certain situations, such as poor weather conditions (heavy rain, fog, snow)

or challenging road conditions. The sensors and cameras used by these systems may have reduced effectiveness or accuracy in such scenarios, which can impact their overall performance.

It's important for drivers to understand both the advantages and limitations of ADAS technologies. They should use these systems as aids to safe driving while maintaining their own attentiveness, judgment, and responsibility on the road. Table 1.1 gives a summary of the advantages and disadvantages of ADAS.

Table 1.1: Advantages and Disadvantages of ADAS

Advantages	Disadvantages
Enhanced Safety	Overreliance and Complacency
Collision Mitigation	False Alerts and Malfunctions
Improved Driver Assistance	High Cost
Increased Convenience	Potential Learning Curve
Potential for Fuel Efficiency	Limited Effectiveness in Certain Conditions

1.2 Lane Departure Warning System

The Lane Departure Warning System (LDWS) is an advanced driver assistance feature that is designed to enhance safety on the road by alerting drivers when their vehicle unintentionally drifts or departs from its lane without using the turn signal. LDWS utilizes sensors or cameras to monitor the position of the vehicle within the lane and provides timely warnings to the driver, prompting them to take corrective action.

LDWS technology recognizes the importance of maintaining proper lane discipline, as unintentional lane departures can be a significant cause of accidents and collisions. By monitoring the vehicle's position relative to the lane markings on the road, LDWS acts as a vigilant co-pilot, providing an additional layer of safety to help prevent potential accidents resulting from driver distraction, drowsiness, or momentary lapses in attention.

When the LDWS sensors or cameras detect that the vehicle is drifting or veering out of its lane without the use of a turn signal, the system generates an alert. The warning

can take the form of an audible sound, visual notification on the instrument cluster or heads-up display, or even a vibration in the steering wheel or seat. This immediate feedback prompts the driver to take corrective action, such as steering back into the proper lane.

It is important to note that LDWS is designed as a warning system and does not actively intervene to steer the vehicle. Its primary goal is to alert and assist the driver in maintaining their intended lane position, ultimately reducing the risk of accidents caused by unintentional lane departures.

While LDWS can be a valuable aid in promoting safer driving practices, it is essential for drivers to remain attentive and not solely rely on the system. LDWS may have limitations, such as reduced effectiveness in adverse weather conditions or on poorly marked roads. Therefore, it is crucial for drivers to always practice good driving habits, including maintaining their focus, using turn signals, and keeping a proper lookout for lane markings and surrounding vehicles.

Overall, LDWS serves as a helpful technology that aims to enhance road safety by providing drivers with timely alerts and reminders to stay within their lanes, ultimately contributing to a safer driving experience for all.

Here's how a typical LDWS works:

1. **Sensors or Cameras:** LDWS relies on sensors or cameras, usually mounted near the rearview mirror or on the front windshield, to monitor the vehicle's position in relation to the lane markings on the road.
2. **Lane Detection:** The system analyzes the input from the sensors or cameras to detect the lane markings, such as painted lines or raised markers. It identifies the position of the vehicle within the lane.
3. **Warning Generation:** If the system detects that the vehicle is drifting or veering out of its lane without the use of a turn signal, it generates a warning to alert the driver. The warning can be in the form of an audible alert, visual warning on the instrument cluster, or vibration in the steering wheel or seat.
4. **Sensitivity and Settings:** LDWS often allows drivers to adjust the sensitivity of the system or turn it on/off according to their preferences. This customization can be useful in adapting the system to different driving conditions or personal preferences.

The working principle of LDWS involves image processing techniques to analyze camera data and detect lane markings. The image processing algorithms used in LDWS

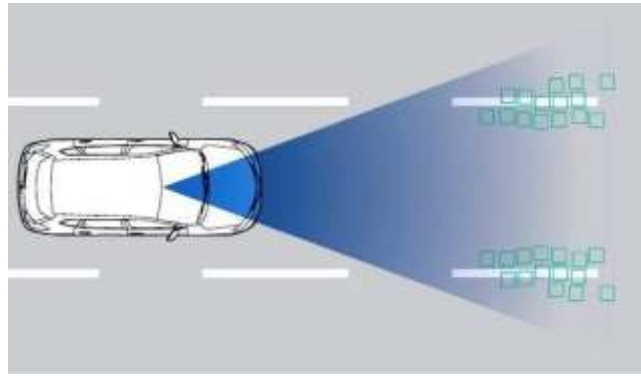


Figure 1.3: An outline of LDWS

are designed to handle various road and lighting conditions, different lane marking styles, and potential challenges like occlusions or noisy image data. These algorithms leverage techniques such as edge detection, feature extraction, pattern recognition, and machine learning to accurately detect and track the lane markings.

It's important to note that while image processing is a fundamental aspect of LDWS, other technologies, such as sensor fusion and decision-making algorithms, may also be employed to enhance the system's performance and reliability. By utilizing image processing techniques, LDWS systems can effectively analyze camera data, interpret the lane markings, and provide timely warnings to assist drivers in maintaining proper lane discipline.

One can activate lane departure warning systems either by pressing a button or it activates automatically when you start the car. The button will show a light indicating that the system is active. The lane departure system detects the lane markings on both straight and curved roads. Moreover, this system doesn't alert the drivers when they respond by turning the steering wheel quickly and keeping their cars in the lane.



Figure 1.4: Deactivation of LDWS when indicator is ON

1.2.1 Advantages and Disadvantages of LDWS

Lane Departure Warning System offer several advantages and disadvantages. Let's explore them:

Advantages of LDWS

1. **Improved Safety:** LDWS enhances road safety by providing a warning to drivers when they unintentionally drift out of their lane. This can help prevent accidents caused by lane departure, such as collisions with other vehicles or objects on the road.
2. **Reduced Fatigue and Distraction:** Long drives or monotonous highways can lead to driver fatigue and distraction, increasing the risk of lane departure. LDWS can help mitigate this by alerting drivers and keeping them focused on the road, reducing the chances of accidents due to lapses in attention.
3. **Early Warning System:** LDWS can detect lane departures early on and provide timely warnings, allowing the driver to take corrective action. This can be especially beneficial in situations where the driver is momentarily distracted or experiences a lapse in concentration.
4. **Customization and Flexibility:** Many LDWS systems offer customizable settings, allowing drivers to adjust the sensitivity of the system or choose the type of warning (audible, visual, or haptic). This flexibility enables drivers to tailor the system to their preferences and driving style.

Disadvantages of LDWS

1. **False Alarms:** LDWS may occasionally generate false alarms, especially in certain driving conditions or when road markings are faded or unclear. These false alarms can be annoying and may potentially lead to driver complacency or disengagement from the system.
2. **Overreliance and Reduced Vigilance:** Drivers who become overly reliant on LDWS may develop a false sense of security, assuming the system will always detect lane departures and intervene.
3. **Limited Effectiveness:** LDWS primarily relies on detecting road markings to determine lane departure. In situations where road markings are absent or inadequate, such as construction zones or rural roads, the system's effectiveness may be compromised. Additionally, LDWS may not be effective in preventing lane departure caused by sudden maneuvers or aggressive driving.
4. **Cost:** The implementation of LDWS adds to the overall cost of a vehicle. While the availability of LDWS has increased in recent years, it may still be more commonly found in higher-end or newer vehicles.

It's important to note that LDWS is a driver assistance system and not a substitute for responsible driving. Drivers should always remain attentive, maintain control of their vehicles, and adhere to traffic laws and road conditions, even when using LDWS. Table 1.2 gives a summary of the advantages and disadvantages of LDWS.

Table 1.2: Advantages and Disadvantages of LDWS

Advantages	Disadvantages
Improved Safety	False Alarms
Reduced Fatigue and Distraction	False Alerts and Malfunctions
Early Warning System	Overreliance and Reduced Vigilance
Customization and Flexibility	Limited Effectiveness
Potential for Fuel Efficiency	Cost

1.3 Image Processing

Image processing is the analysis and manipulation of visual information captured by digital images, such as photographs or video frames. Image processing techniques can be used to extract useful information from images, enhance image quality, and perform various other tasks related to image analysis and recognition.

In the context of road safety, image processing can be used to analyze visual information captured by cameras or other sensors to detect and recognize the lanes on the road. There are several image pre-processing and processing techniques that can be used to analyze and manipulate digital images. Some of the common techniques used in image processing are discussed here.

1. **Image Enhancement:** This technique is used to improve the visual quality of an image by adjusting parameters such as brightness, contrast, or color balance.
2. **Image resizing:** It is the process of changing the size of a digital image, typically by reducing or enlarging its dimensions. Resizing an image can be useful for a variety of purposes, such as reducing the file size for faster loading times, fitting an image into a specific layout or design, or creating a smaller version of an image for use on mobile devices or social media.
3. **Image normalization:** It is a pre-processing technique used in computer vision and image processing to adjust the pixel values of an image to a common scale or range.

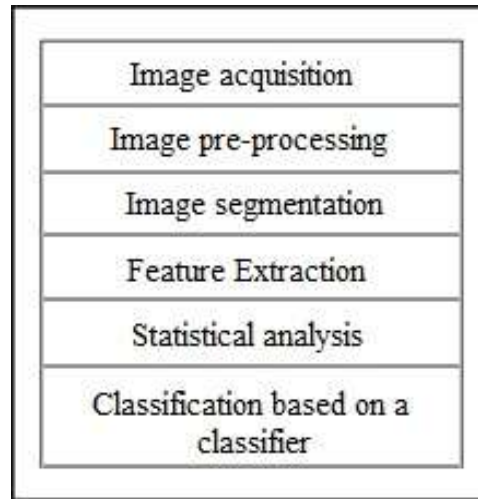


Figure 1.5: Steps in Digital Image Processing

4. Image compression is the process of reducing the size of an image file without significantly reducing its quality or visual fidelity. The goal of image compression is to reduce the amount of storage space required to store an image, as well as to reduce the amount of time it takes to transmit an image over a network.
5. Image recognition, also known as image classification or image analysis, is the process of identifying and categorizing objects or patterns within digital images.

LDWS utilizes image processing techniques to detect and warn drivers when their vehicle unintentionally departs from its current lane. The specific image processing technique commonly used in LDWS is called “Lane Detection”. Lane Detection typically involves the following steps:

1. Image Acquisition: The system captures video frames of the road ahead using a camera mounted on the vehicle.
2. Pre-processing: The captured images may undergo pre-processing to enhance their quality and remove noise. This step may include operations like resizing, noise reduction, and contrast adjustment.
3. Color Space Conversion: The images are often converted from the RGB color space to a different color space, such as grayscale or HSL (Hue, Saturation, Lightness), to simplify subsequent processing steps.
4. Edge Detection: The system applies edge detection algorithms, such as Canny edge detection, to identify the edges of lane markings in the image. This helps in distinguishing the lanes from the surrounding road environment.
5. Hough Transform: The Hough transform is used to detect and extract lines from the edge-detected image. This step helps in finding the prominent lines representing the lane markings.
6. Lane Modeling: The extracted lines are then analyzed and processed to model and estimate the lanes. Various techniques, such as curve fitting or polynomial regression, can be employed to create a mathematical representation of the lanes.

7. Lane Departure Warning: Once the lanes are modeled, the system compares the vehicle's position within the lanes with respect to the estimated lane positions. If the vehicle deviates significantly from the lane, the LDWS triggers a warning, such as audible alerts, visual indicators, or haptic feedback, to alert the driver.

1.3.1 Edge Detection

Canny edge detection is an image processing technique commonly used in LDWS for detecting the edges of lane markings in captured images or video frames. It helps identify the boundaries of lanes, which are crucial for lane detection and tracking algorithms. Here's how Canny edge detection works in LDWS:

- **Grayscale Conversion:** The input image, typically in RGB color space, is first converted to grayscale. This simplifies subsequent edge detection steps by reducing the image to a single channel.
- **Noise Reduction:** Before applying edge detection, it is common to reduce noise in the image to obtain cleaner edge results. This can be achieved by applying a Gaussian blur filter to smooth out pixel variations caused by noise while preserving important edge information.
- **Gradient Calculation:** Canny edge detection utilizes the concept of gradients to identify edges. The gradients represent the rate of change of pixel intensity across the image. In this step, the gradients are calculated using edge detection kernels, such as the Sobel operator, in both the horizontal and vertical directions.
- **Non-maximum Suppression:** The goal of this step is to refine the detected edges by thinning them to a single pixel width. The algorithm examines the gradient magnitudes and suppresses (sets to zero) the gradients that are not local maxima in their respective directions. This ensures that only the most significant edges are retained.
- **Double Thresholding:** Two threshold values, a high threshold and a low threshold, are used to classify the edges into strong and weak edges. Strong edges are likely to be part of the lane markings, while weak edges may contain noise or less significant features. Pixels with gradient magnitudes above the high threshold are considered strong edges, while those below the low threshold are discarded. Pixels with gradient magnitudes between the two thresholds are considered weak edges.
- **Edge Tracking by Hysteresis:** Weak edges that are connected to strong edges are retained, while isolated weak edges are discarded. This is achieved by performing edge tracking using hysteresis. Starting from a strong edge pixel, the algorithm follows the weak edges that are connected, considering them part of the edge. This process ensures that only continuous edges, likely corresponding to lane markings, are preserved.

The result of Canny edge detection is a binary image where the detected edges are represented by white pixels on a black background. These edges can then be further processed and analyzed in subsequent steps of the LDWS algorithm, such as the Hough transform, to extract lane lines and estimate the vehicle's position relative to the lanes.

1.3.2 Hough Transform

The Hough transform is an image processing technique commonly used in LDWS (Lane Departure Warning System) to detect and extract lines from the edge-detected image obtained through techniques like Canny edge detection. It is particularly useful for identifying lane markings and estimating their parameters, such as slope and intercept. Here's how the Hough transform works in LDWS:

- **Edge Detection:** Before applying the Hough transform, the input image is typically processed to detect edges using techniques like Canny edge detection. This step helps identify the boundaries of the lane markings.
- **Parameter Space:** The Hough transform represents lines in a parameter space instead of the spatial domain. In the case of the Hough transform for lines, the parameter space consists of two parameters: slope (m) and intercept (b) of the line equation ($y = mx + b$). Each point in the parameter space corresponds to a possible line in the image.
- **Voting:** For each edge pixel detected in the edge image, the Hough transform algorithm votes in the parameter space to determine the lines that could have generated that pixel. It iterates through the edge pixels and, for each pixel, increments the corresponding bins in the parameter space that satisfy the line equation.
- **Accumulator Thresholding:** After the voting process, the parameter space represents the strength of different line hypotheses. To extract the most significant lines, an accumulator threshold is applied to identify the bins in the parameter space that have accumulated enough votes. These bins correspond to the parameters of the detected lines.
- **Line Extraction:** The bins above the accumulator threshold are considered as lines detected in the image. The parameter values (slope and intercept) associated with these bins are converted back to spatial coordinates, providing the coordinates of the detected lines in the image.
- **Post-processing:** The extracted lines can undergo additional post-processing steps to refine their representation and remove any outliers or noise. Techniques such as line merging, line filtering, or line extrapolation can be applied to improve the accuracy and continuity of the detected lane markings.

By applying the Hough transform, the LDWS algorithm can identify and extract lines representing the lane markings in the image. These lines serve as a basis for

further analysis, such as lane modeling, estimation of lane curvature, and determining the vehicle's position with respect to the lanes. The Hough transform helps robustly detect lane lines, even in the presence of noise, gaps, or partial occlusions, making it a valuable component of LDWS systems. In general the straight line equation is given by $y = mx + c$ can be represented as point (c, m) in the parameter space or Cartesian co-ordinate system. For the Hough transform we convert the equation to polar co-ordinates i.e. in terms of rho and theta.

Kalman Fiter

The Kalman filter is an algorithm used for estimating the state of a system by combining measurements with predictions based on a mathematical model. It is particularly useful in situations where there is uncertainty or noise in the measurements and where the system's state evolves over time. In the context of LDWS, the Kalman filter can be employed to improve the accuracy and reliability of lane tracking. Here's how the Kalman filter can be used in LDWS:

- **State Estimation:** The Kalman filter can estimate the current state of the vehicle with respect to the lanes, considering variables such as lateral position, heading angle, and velocity. The initial state is obtained from the lane detection algorithms or other sensor inputs.
- **Measurement Fusion:** LDWS typically utilizes multiple sensors to gather information about the vehicle's position and lane markings. These sensors can include cameras, GPS, inertial sensors, and more. The Kalman filter combines the measurements from these different sensors, taking into account their respective uncertainties, to provide a more accurate and robust estimate of the vehicle's state.
- **Smoothing and Filtering:** The Kalman filter not only estimates the current state but also provides a smoothed trajectory of the vehicle over time. It achieves this by incorporating information from past measurements and predictions, allowing for a more consistent and stable estimate of the lane position.
- **Prediction:** The Kalman filter utilizes a mathematical model of the vehicle's dynamics to predict its future state based on the current estimated state. This prediction takes into account factors such as the vehicle's speed, acceleration, and steering angle. By combining the prediction with the actual measurements, the Kalman filter can adapt to changes in the lane environment and compensate for measurement errors.
- **Error Correction:** The Kalman filter dynamically adjusts the weights given to the measurements and predictions based on their respective uncertainties. It provides a mechanism to correct errors and minimize the impact of noisy or erroneous measurements, resulting in improved accuracy and reliability of the LDWS.

Euclidean Distance

In the context of LDWS, the Euclidean distance method can be used as a mathematical measure to determine the distance between the detected lane markings and the current position of the vehicle. This method is commonly employed in LDWS to assess the deviation of the vehicle from the lane and trigger appropriate warnings or corrective actions.

- Once the lane markings are detected using image processing techniques, such as the Canny edge detection and the Hough transform, the Euclidean distance method can be applied. The method calculates the straight-line distance between the vehicle's position and the lane markings in the two-dimensional image plane. By comparing this distance with a predefined threshold, the LDWS can determine if the vehicle has deviated from the lane.
- If the calculated distance exceeds the threshold, it indicates a potential lane departure, and the LDWS can then issue warnings to alert the driver. These warnings can be in the form of audible alerts, visual indicators on the dashboard, or haptic feedback to get the driver's attention and prompt corrective action.

1.4 Proposed System

The proposed system of LDWS helps in detecting and locating the lane of road continuously. If the lane is found broken the vehicle (lane gets departed) system gives an alert of a beep sound along with the message that the lane has been broken. In real-time this is implemented using camera sensors attached to the car and a display device integrated with an audio system inside the vehicle.



Figure 1.6: Proposed system of LDWS

CHAPTER 2

LITERATURE SURVEY

Performing a literature study for a project is an important step in its research and presenting findings. Here are the details of the journals that we had studied from ScienceDirect, Sustainability journals and IJRSI journals during the project work.

2.1 Review on Lane Detection and Tracking Algorithms of Advanced Driver Assistance System

The review paper titled ‘Review on Lane Detection and Tracking Algorithms of Advanced Driver Assistance System’ [1] deals with three types of approaches for lane detection which are: learning based approach, features-based approach, and model-based approach. Specifically, the review identifies and classifies the existing lane detection and tracking algorithms under three themes: features-based, learning-based and model-based, which provides a systematic approach towards understanding the key characteristics of lane detection and tracking algorithms in the literature. A comparison of the different sensors used in ADAS is presented first. It is then followed up with an in-depth review of algorithms used for lane detection and tracking.

In this paper, three types of sensors have been identified that are used in the ADAS, which are, Light Amplification by Stimulated Emission of Radiation (LASER) based sensors, Radio Detection And Ranging (RADAR) based sensors and vision-based sensors. SWOT analysis of these sensors are also presented in this paper.

The feature-based approach uses edges and local visual characteristics of interest, such as gradient, colour, brightness, texture, orientation, and variations, which are relatively insensitive to road shapes but sensitive to illumination effects. The model-based approaches apply global road models to fit low levels of features that are more robust against illumination effects, but they are sensitive to road shapes. The geometric parameters are used in the model-based approach for lane detection. The learning-based approach consists of two stages training and classification.

2.2 Robust and real-time multi-lane and single lane detection in Indian highway scenarios

The research paper titled ‘Robust and real-time multi-lane and single lane detection in Indian highway scenarios’ [2] dealt with vision-based lane detection frameworks which are normally planned dependent on image processing procedures inside comparative structures.

Two different sub-algorithms make up the suggested algorithm. The Fuzzy Noise Reduction Filter (FNRF) is the first sub-algorithm, which reduces noise and smooth’s the sequences of images acquired by the camera. The second sub-algorithm combines the Hough Transform (HT) technique with a competent zone of interest to recognize lanes in both normal and difficult situations.

The paper also suggested a CNN-based regression technique for identifying multiple lanes. Pre-processing, post-processing, and lane recognition are the three main components of this architecture. These components imitate human abilities to recognize street lanes by perceiving with the eyes, analyzing with the mind, and then providing an analyzed result.

2.3 Image Processing based Lane Departure Warning System Using Hough Transform and Euclidean Distance

The research paper titled ‘Image Processing based Lane Departure Warning System Using Hough Transform and Euclidean Distance’ [3] explains the methodologies used in lane identification for developing LDWS. In lane departure warning system, the lane detection is the initial step to be taken. There are two types of methodologies used in lane identification: the elements based methodology and the model based methodology. The elements based methodology detects the lane from the images of roads by detecting the low level elements such as lane edges etc. This methodology requires all round painted lanes or solid lane edges, otherwise it cannot detect it.

This methodology may experience the ill effects of impediment or clamor. The model based methodology use geometric use geometric parameter such as assuming the shape of the lane can be presented by straight line or curve.

In this paper, the piecewise stretching function is used to set the color contrast of the road lane images as a method of image pre-processing. This function makes the roads images clear enough to detect the lanes. The Region of Interest (ROI) is set to lessen the computational time of images. Followed by image processing techniques like edge detection and Hough transform.

2.4 Lane Departure System Design using with IR Camera for Night-time Road Conditions

The research paper titled ‘Lane Departure System Design using with IR Camera for Night-time Road Conditions’ [4] consists of a research and development of an algorithm for automotive systems to detect the lane departure at night-time. Lane detection at night-time is an important issue in driving assistance systems.

A head-up display or heads-up display also known as a HUD is any transparent display that presents data without requiring users to look away from their usual viewpoints, a form of augmented reality (AR).

In this paper, the authors proposed an efficient, real time, and robust algorithm for detecting lanes at night-time. With recent advances of Head-up Display technology in cars, Augmented Reality becomes interesting in supporting the driving task to guide a driver’s attention.

2.5 Lane Departure Warning System

The journal paper titled ‘Lane Departure Warning System’ [5] gives a clear understanding of the implementation of the model by make use of image processing techniques. The cameras installed in the vehicle will capture images of lanes on roads and image processing techniques can be used for preprocessing. 1-D or 2-D Gaussian smoothing

can be used to reduce noise in the captured image. Edge detection methods can be used to filter out unnecessary background information and to detect edges of lanes.

Line gradient estimation method is one method to detect lane. Another method for lane detection is Hough transform which requires less computational time to detect a lane marking over the entire frame when compared to the former. The Gradient method cannot distinguish false intensity bumps caused by leading vehicles and road markings/textures from weak lane markings. Various platforms like FPGA, ARM, and DSP can be used to implement LDWS, each of these has its own advantages and disadvantages which are presented by the authors.

FPGA is more advantageous compared to other commercially available hardware platforms such as microprocessors, DSP due to its features such as re-configurability, flexibility. It takes less time for computation compared to others, due to its parallel nature. Real time processing of image is difficult with the help of FPGA due to resource constraints. Digital signal processors (DSP) supports real time processing. Thus, DSP based embedded system can be used to implement vision based LDWS. LDWS can be built using Advanced Risc Machines (ARM). This gives portable system with properties compact size, low power consumption and low cost, can widely be deployed to various popular consumer devices.

CHAPTER 3

OBJECTIVES

The objective of a Lane Departure Warning System (LDWS) project is to enhance road safety by providing drivers with a warning when their vehicle unintentionally drifts out of its current lane. The primary goal is to prevent or mitigate potential accidents caused by lane departure incidents, which can occur due to driver inattention, drowsiness, distraction, or other factors.

Here we prevent the occurrence of road accidents in today's world that focuses on preventing accidents by the use of image processing techniques. For that we developed a system (LDWS) that detect the departure of lane by the vehicle which are driven by the driver happening in real-time, thereby giving timely alert to the drivers by a beep audio sound as an alert, so that the can correctly maintain the proper lane while driving. This can avoid road accidents up to a certain limit. Figure 3.1 shows the basic block diagram of LDWS.

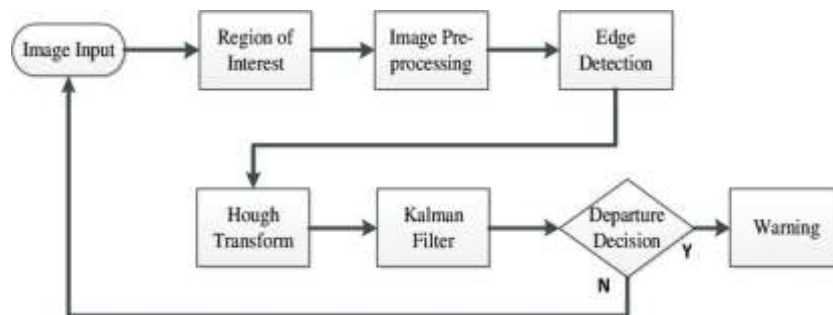


Figure 3.1: Basic block diagram of LDWS

3.1 Project Plan

The purpose of project plan is to provide a clear understanding of the project's objectives, scope, and requirements to provide a framework for managing and executing the project. Our project development is divide into different sections like the literature survey, study of methodologies, study and investigation of system design and real time implementation using laptop camera, and a project report.

3.1.1 Prerequisite for project

Following are the prerequisites needed to be studied and understand in order to complete the project:

1. Basic knowledge in Python programming
2. Understand different image processing techniques
3. Study of different tools like Python IDLE, Anaconda prompt and Sublime text3

Figure 3.2 shows a road-map plan of the project that outlines the steps needed to be taken in order to successfully complete the project.

Duration in Weeks. Name of Task.	Week 1	Week 3	Week 5	Week 7	Week 9	Week 10	Week 11	Week 12
Literature Study								
Study of Methodology								
Study and Investigation of System Design								
Study and Investigation of Real-Time Implementation								
Report Writing								

Figure 3.2: Road Map

CHAPTER 4

DESIGN AND SPECIFICATIONS

4.1 Design

The design in our project encompasses a wide range of activities, from creating rough sketches and wireframes of the system to outlining the user experience and user interface design. It can also involve designing the project's architecture, data models, and software and hardware specifications. Following are the software specifications used in our project:

4.1.1 Software Specifications

Software specifications typically include a description of the user interface, inputs and outputs, data structures, algorithms, performance requirements, and any other relevant aspects of the software system. They also include details about the software's functionality, its interaction with other systems, and any constraints or limitations that must be considered during development. The hardware and software requirements used here are:

- i. Hardware requirements
 - Processor: i5 or i7
 - RAM: 8GB
 - Hard Disk: 500GB or above
- ii. Software Requirements
 - Tools: Python IDLE, Anaconda, Sublime Text3
 - Python: version 3.6
 - Operating System: Windows 10

Hardware requirements for the development of intelligence part

The processors i5 and i7 are both Intel processor families commonly used in computers. The main difference between the two is their performance level and the number of cores they have. i5 processors typically have four cores and are generally used in computers for everyday use, such as browsing the internet and using productivity software. In summary, i5 processors are suitable for general use and for this project we use i5, while i7 processors are better for heavy-duty tasks that require more processing power.

RAM (Random Access Memory) is a type of computer memory that is used for temporarily storing data and code that is currently being used by the computer's processor. It is an important component in determining a computer's performance and responsiveness. A minimum of 8GB of RAM is generally recommended for most modern computers, especially for those running resource-intensive applications like virtualization. 8GB of RAM is usually sufficient for multitasking, browsing the internet and using productivity software.

A hard disk drive (HDD) is a type of storage device used in computers for storing data. The capacity of a hard disk is usually measured in gigabytes (GB) or terabytes (TB). A minimum of 500GB of hard disk capacity is recommended for most modern computers, especially for those that are used for general purposes such as browsing the internet, storing files, and running productivity software. This capacity is usually sufficient for storing a moderate amount of data-set documents, photos and videos footages of roads.

Software requirements for the development of intelligence part

Python IDLE: Python IDLE (Integrated Development and Learning Environment) is an integrated development environment for Python programming language. It provides an interactive shell, a code editor with syntax highlighting and autocompletion, and debugging tools to help developers write, test, and debug Python code.

Anaconda3 (64-bit): Anaconda is a popular open-source distribution of Python and R programming languages for data science and machine learning tasks. It comes with a collection of pre-installed libraries and tools that make it easier for data scientists and machine learning engineers to work with large datasets, perform data analysis, and build

machine learning models.

Sublime Text3: Sublime Text is a popular code editor that provides a sleek and minimalist user interface and powerful editing features. It is highly customizable and supports syntax highlighting, autocompletion, multiple selections, and other productivity-boosting features that make it a favorite among developers.

Python 3.6 is used here which support many popular Python libraries and frameworks. Some of them are not inbuilt in python 3.6, so we use `'pip install <name>'`. It is a command in Python that is used to install packages from the Python Package Index (PyPI) or other indexes. In Python, a package is a collection of related modules that can be imported together, while a library is a collection of functions, modules, or objects that can be used to perform specific tasks. Following libraries are used in our coding:

Table 4.1: Libraries used in Python

Libraries	Description
cv2	OpenCV, this library is used for image and video processing. It provides various functions for reading, manipulating, and displaying images and videos.
np	Numerical Python, this library is used for numerical computations and array operations. It provides a powerful N-dimensional array object that allows efficient manipulation of large arrays.
winsound	This library is specific to Windows operating system and is used for playing sounds. In the code, it is used to play a beep sound as an alert in case of lane departure detection.

CHAPTER 5

METHODOLOGY AND WORKING

Methodology includes the overall approach, strategy and set of methods used to conduct the project. This involves a systematic and organized way of approaching a problem, which helps to ensure that the project is carried out efficiently, effectively, and with the desired outcomes. Working of the project is the process of implementing this methodology and carrying out the actual work needed to complete the work. The LDWS, an intelligent traffic monitoring systems (ADAS) are designed to collect and analyze data related to traffic patterns and conditions in real-time. The system uses advanced technologies such as computer vision, machine learning, and data analytics to monitor and manage traffic on roads, highways, and bridges.

The LDWS project developed here uses image processing detect and continuously monitor lanes on road and if found departed an alert with a beep sound is given by the audio system to the drivers so that they can correct the departed lane from looking at the display device which records the live video from the cameras mounted to vehicle.

Methodology and Working of LDWS typically involves the following steps:

- i. Data collection: The system collects data from the camera sensors installed on the vehicle.
- ii. Data processing: The collected data is processed using advanced computer vision and image processing techniques to extract relevant information like the ROI.
- iii. Data analysis: The system analyzes the processed data to identify any departure in the lane and thereby giving timely beep alert sound as an indication of lane departure.
- iv. Decision-making: Based on the analysis results, the drivers can makes decisions to correct the departed lane.

The system continuously monitors the lanes on road in real-time and provides instant alerts in case of any unusual events like a departure is occurred. This uses image processing techniques like edge detection (Canny Edge Detection) and Hough Transform for detecting lanes. If the detection of lanes is failed, then this indicates that

the vehicle has crossed the lanes indicating lane departure, thereby giving timely beep sounds as an alert to the drivers so that they can maintain the lanes correctly, and prevent any occurrence of accidents.

5.1 LDWS as an Accident Prevention Model

This model of LDWS is implemented as a software part that also works in real-time using the laptop front camera.

A Python code (LDWS.py) sets up a video capture using OpenCV to read frames from a camera feed. It applies various image processing techniques such as Gaussian blur, Canny edge detection, and Hough transform to detect and track lanes on the road. It defines a region of interest (ROI) for lane detection and uses a line detection algorithm to identify lane lines. If no lanes are detected for a certain number of frames, it triggers a lane departure alert by displaying a message on the frame and playing a beep sound. The processed frames are displayed in a window until the user presses 'q' to quit. Overall, this code implements a Lane Departure Warning System (LDWS) using computer vision techniques to detect potential lane departure events in real-time video streams.

The code starts by importing necessary libraries like cv2 for computer vision operations, numpy for numerical computations, and winsound for playing a sound alert. Next, it sets up video capture by creating a VideoCapture object. In this code, it captures frames from a live camera feed using argument 0 as the camera index. The code then defines the region of interest (ROI) for lane detection. The ROI is specified as a polygon using coordinates that define a trapezoidal shape in the lower portion of the frame. The kernel size is defined to determine the size of the Gaussian blur kernel used to reduce noise in the grayscale image.

Parameters for Canny edge detection and Hough transform are defined. These parameters control the edge detection and line detection algorithms. Canny edge detection is applied to detect edges in the image, while the Hough transform is used to detect lines in the edges. Lane departure alert parameters are defined. The alert_count variable keeps track of consecutive frames without detected lanes, and the alert_threshold specifies the number of frames required to trigger a lane departure alert.

The code defines the `process_frame` function, which takes a frame as input and performs lane detection and alerting. The function converts the frame to grayscale, applies Gaussian blur to reduce noise, and performs Canny edge detection. It then applies a mask to focus on the region of interest defined earlier. The Hough transform is applied to detect lines in the masked edges, and the detected lines are drawn on a separate line image. The line image is combined with the original frame using the `cv2.addWeighted` function. The code checks for lane departure by analyzing the detected lines. If no lines are detected, the `alert_count` is incremented, otherwise, it is reset. If the `alert_count` exceeds the `alert_threshold`, a lane departure alert is displayed on the frame, and a beep sound is played using `winsound.Beep`.

The code enters a loop where it reads frames from the video capture. For each frame, it calls the `process_frame` function and displays the processed frame with lane detection. If the 'q' key is pressed, the loop is broken and the program exits. Finally, the video capture is released and the windows are closed. Figure 5.1 and 5.2 shows the steps and flow chart for the working of LDWS.

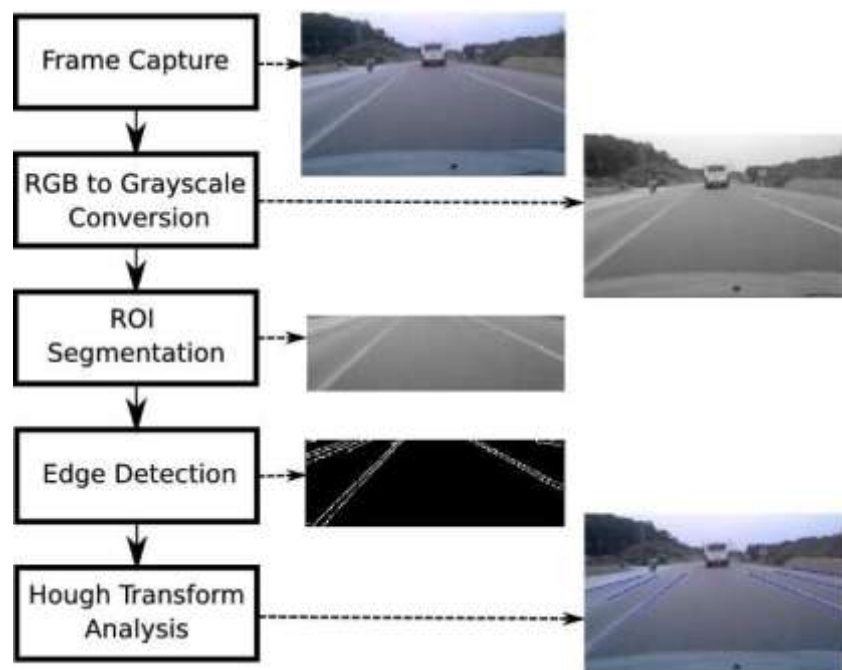


Figure 5.1: Steps showing the working of LDWS

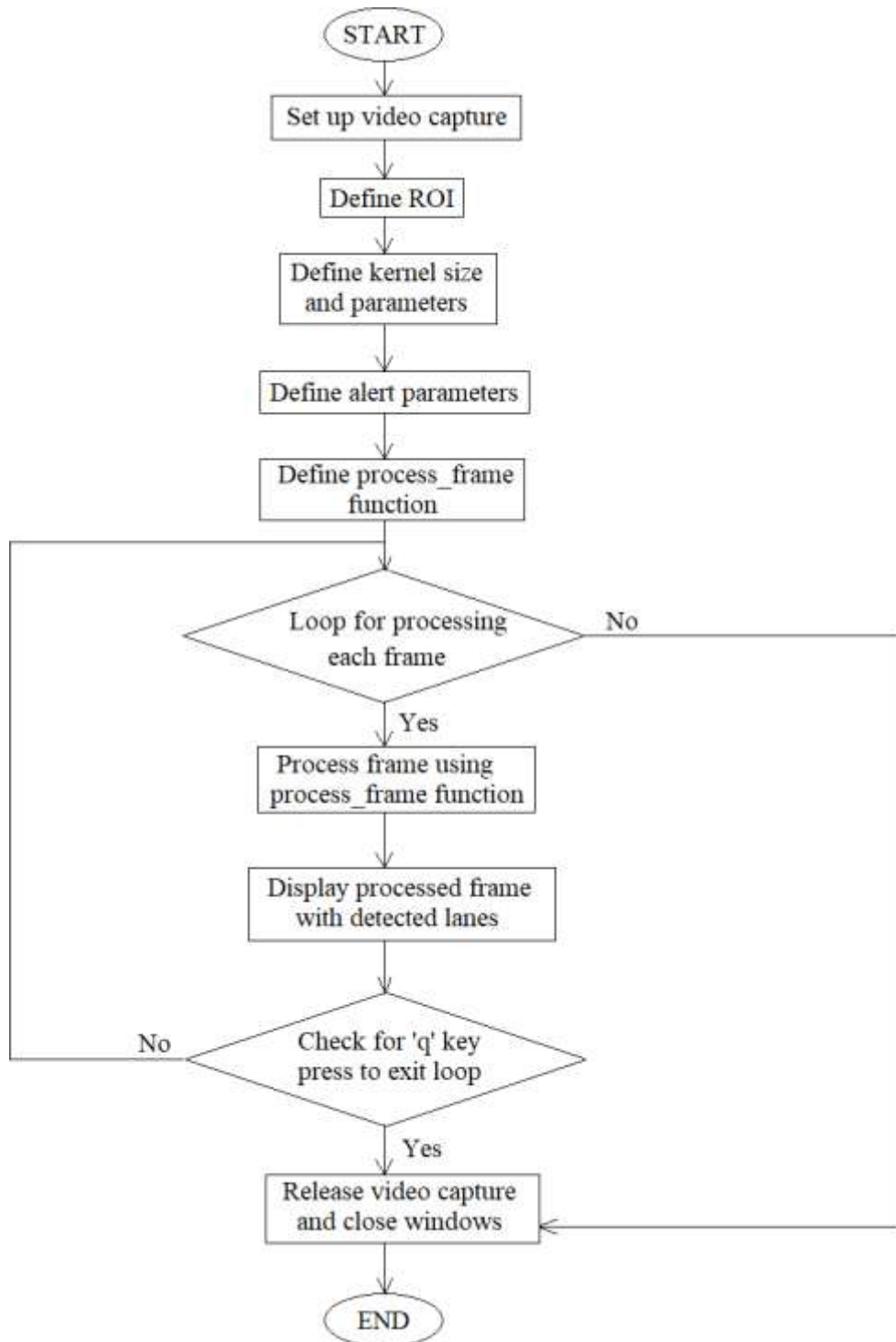


Figure 5.2: Flow chart showing the working of LDWS

CHAPTER 6

RESULTS AND DISCUSSIONS

6.1 LDWS in Real-time Detection

On detection of a lane departure, alerts are received by the drivers as a beep sound indicating that there is a lane departure has been detected by the system. The decision of correcting and maintain a proper lane while driving the vehicle can be done by the driver in order to prevent any occurrences of an accident.

This system is developed using a python code (LDWS.py). Figure 6.1 shows the console along with its result using a dummy road and car.



Figure 6.1: LDWS- (a) Normal (b) Departure detected and produced a beep sound

6.2 Investigation Results

The real-time results suggests an Inverse projective mapping (IPM) to create a bird's-eye view of the road- a Hough transform for detecting candidate lane and Kalman filter track the lane. The road image is converted to grayscale form followed by temporal blurring. The application of IPM makes the image provide a bird's eye view. The lanes

are detected by identifying the pair of parallel lines which are separated by a distance. The IPM images are converted to binary, and a Hough transform is performed on the binary image and then divided into two halves. To determine the center of the line, the one-dimensional matched filter is applied to each sample. The pixel with a large correlation that exceeds the threshold is selected as the center of the lane. The Kalman filter is used to track the lane, which takes the lane orientation and difference between the current and previous frames. A firewire camera is used to capture the image of the road. The performance of the proposed algorithm provides better accuracy under the isolated highway and metro highway, and the accuracy is in the range of 86% on city roads. The improved performance is due to the usage of the Kalman filter to track the lane.

The captured image in RGB format is converted to gray code followed by binary image conversion. The purpose of binary image conversion is to remove the shadows in the captured image. The lanes in the image are detected using the canny algorithm by the feature named color. The direction and intensity are determined by removing the noise using the gaussian filter. The images are smoothened by applying a median filter. The lanes in the image are considered as the region of interest, and Hough transform is applied to confirm the accuracy of the lanes in the region of interest. The experiment is to be performed during the daytime. The results will be more than 93% accurate. We have surveyed different algorithms for lane detection and lane tracking, such as gradient algorithm, Hough transforms (HT), RANSAC (Random Sample Consensus) and deep learning (CNN). We have gone through various problems to be dealt with during lane detection like worn lanes, lane curvature, emerging lanes, shadow and light problem. RANSAC algorithm provides solution for most of these problems, thus it is more beneficial compared to others. More over this accuracy can be improved by using learning based approaches like deep learning techniques. The system can achieve an accuracy of 97% in lane detection if we are using machine learning techniques.

According to our study and investigation of literature review papers, following table gives a comparison results of different algorithms used for tracking lanes for different LDWS projects.

Table 6.1: Comparison Results

	Gradient Approach	Hough Transform	RANSAC	Deep Learning Algorithm
Methodology	Based on image gradients and edge detection	Accumulates votes in parameter space for line detection	Iteratively fits models to data and handles outliers	Utilizes neural networks to learn from labeled data
Lane Boundary Detection	Relies on gradient-based edge detection	Detects lines or curves as lane boundaries	Fits lines or curves to lane boundaries	Learns lane features and representations from data
Robustness	May struggle in complex or noisy road conditions	Can handle noise and outliers with proper parameter tuning	Robust to outliers and noise in the data	Robust to varying road conditions and lane markings
Computational Complexity	Relatively simple and computationally efficient	Higher computational complexity	Moderate computational complexity	Higher computational complexity
Adaptability	Limited adaptability to varying road conditions	Can handle different types of lane boundaries	Adaptable to noise and outliers	Adaptable to diverse road conditions and lane markings
Real-time Performance	Achievable with optimized implementations	Real-time performance may be challenging	Real-time performance achievable	Real-time performance achievable
Training Data Requirement	Does not require labeled training data	Does not require labeled training data	Does not require labeled training data	Requires a large amount of labeled training data

CHAPTER 7

CONCLUSION

In conclusion, the Lane Departure Warning System (LDWS) is a valuable driver assistance feature that contributes to enhanced road safety and accident prevention. LDWS provides early warnings to drivers when they unintentionally drift out of their lanes, helping to mitigate the risks associated with driver fatigue, distraction, and lapses in attention. With its customizable settings, LDWS offers flexibility to adapt to individual preferences and driving styles. However, it is essential to acknowledge the limitations of LDWS, including the potential for false alarms, the risk of overreliance leading to reduced driver vigilance, and its effectiveness in specific driving conditions. While LDWS can be a beneficial addition to vehicles, it should always be complemented by responsible driving practices and adherence to traffic laws. As technology advances, continued improvements in LDWS and its integration with other driver assistance systems hold promise for further enhancing road safety in the future.

The designed intelligence part model was successfully able to detect and give warning sign like the beep sound when the lane departures by the vehicle. This project successfully implemented an LDWS using image processing techniques such as Canny edge detection and the Hough transform, combined with the programming language Python. The developed LDWS demonstrated its capability to detect lane departures by analyzing the edges of lane markings and providing timely warnings to drivers. By leveraging image processing algorithms, this system showcased the potential of utilizing computer vision techniques to enhance road safety and assist drivers in maintaining lane discipline. Future advancements in this area hold the promise of further improving the accuracy and effectiveness of LDWS, ultimately contributing to a safer driving experience for all.

CHAPTER 8

FUTURE SCOPE

The Lane Departure Warning System (LDWS) project has several potential future directions for development and enhancement. Firstly, integrating lane tracking algorithms can enable real-time feedback to drivers, assisting them in maintaining their position within the lane. This can be coupled with control mechanisms to automatically correct the vehicle's trajectory in case of lane deviation, thereby further improving lane-keeping capabilities.

Secondly, exploring advanced image processing techniques, such as deep learning-based approaches, can enhance the accuracy and robustness of the LDWS. These techniques can handle complex road scenarios and varying lighting conditions more effectively, ensuring reliable lane departure detection. Additionally, integrating road sign recognition capabilities into the LDWS can provide valuable information to drivers, including speed limits, stop signs, and construction zone warnings. By extending the system's functionality to include autonomous driving systems, the LDWS can contribute to enhancing the capabilities of autonomous vehicles in lane-keeping and overall driving safety.

Furthermore, real-time driver monitoring can be incorporated into the LDWS by utilizing technologies like facial recognition, eye tracking, and fatigue detection. This approach enables the system to adapt warnings and alerts based on the driver's state, increasing safety and reducing the risk of accidents caused by driver fatigue or distraction. Additionally, extensive testing and validation across diverse road conditions, lighting scenarios, and vehicle types will be crucial to ensure the system's reliability and effectiveness. Comparative evaluations with existing commercial LDWS solutions can provide valuable insights and opportunities for improvement, leading to advancements in driver assistance systems and road safety.

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APPENDIX

This Appendix explains the steps and code undertaken for the development of an LDWS of our project.

1. To use libraries like cv, np and winsound, you may need to install them separately. The OpenCV library (cv2) can be installed using `'pip install opencv-python'`, NumPy (np) can be installed with `'pip install numpy'`, and winsound is typically included by default in Python installations on Windows systems.
2. Open the terminal and change the directory to which the python file is present using the command `'cd <path name>'`
3. Run the python code using the command `'python LDWS.py'`. Now the front camera of the laptop gets activated for detecting lanes on road, giving a beep sound if the vehicle crosses the lanes.

CODE FOR LANE DEPARTURE WARNING SYSTEM

#LDWS.py

```
import cv2
import numpy as np
import winsound

# set up video capture
cap = cv2.VideoCapture(0)

# define region of interest (ROI)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
roi = np.array([[(0, height), (0.45 * width, 0.6 * height), (0.55 * width, 0.6 * height),
(width, height)]] , dtype=np.int32)

# define kernel size for image processing
```

```

kernel_size = 5

# define parameters for Canny edge detection and Hough transform
canny_low_threshold = 50
canny_high_threshold = 150
hough_rho = 1
hough_theta = np.pi/180
hough_threshold = 50
hough_min_line_length = 100
hough_max_line_gap = 50

# define lane departure alert parameters
alert_count = 0
alert_threshold = 5

# define function for processing each frame
def process_frame(frame):
    global alert_count

    # convert frame to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # apply Gaussian blur to reduce noise
    blur = cv2.GaussianBlur(gray, (kernel_size, kernel_size), 0)

    # apply Canny edge detection
    edges = cv2.Canny(blur, canny_low_threshold, canny_high_threshold)

    # apply ROI mask
    mask = np.zeros_like(edges)
    cv2.fillPoly(mask, roi, 255)
    masked_edges = cv2.bitwise_and(edges, mask)

    # apply Hough transform to detect lines

```

```

lines = cv2.HoughLinesP(masked_edges, hough_rho, hough_theta, hough_threshold,
np.array([]), hough_min_line_length, hough_max_line_gap)

# draw detected lines on original frame
line_image = np.zeros((height, width, 3), dtype=np.uint8)
if lines is not None:
    for line in lines:
        x1, y1, x2, y2 = line[0]
        cv2.line(line_image, (x1, y1), (x2, y2), (0, 0, 255), 2)

# combine line image with original frame
result = cv2.addWeighted(frame, 0.8, line_image, 1, 0)

# check for lane departure
if lines is not None and len(lines) > 0:
    alert_count = 0 reset the alert count if lanes are detected
else:
    alert_count += 1 increment the alert count if no lanes are detected

# display alert if lane departure is detected
if alert_count >= alert_threshold:
    cv2.putText(result, 'Lane Departure Detected!', (int(width*0.4), int(height*0.9)),
cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)
winsound.Beep(1000, 500) plays a beep sound for 500ms

# return processed frame
return result

# process each frame of video
while True:
    ret, frame = cap.read()
    if ret:
        result = process_frame(frame)
        cv2.imshow('LDWS', result)

```

```
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break
        else:
            break

# release video capture and close windows
cap.release()
cv2.destroyAllWindows()
```